Workflow

---

1. **Layer 1: Pre-LLM entity masking**
    - The system uses regular expressions to immediately detect the account number "8877665544" in the employee's message.
    - It swaps the number with a placeholder, transforming the message to *"Check loan default for user account <ID_1>"*.
    - The real number is locked in a session-scoped vault so the AI never sees it, preventing the model from leaking or transposing it.
2. **Layer 2: Constrained intent extraction**
    - The AI reads the masked prompt and selects the most relevant action from a curated, safe menu of 30 to 50 business intents.
    - It outputs a strict JSON frame identifying the intent (e.g., check_loan_status) and notes that <ID_1> has a role_hint of "account".
    - The system treats this role_hint strictly as a guess and does not trust it for execution.
3. **Layer 3: Entity resolver**
    - The system pulls the real number from the vault and runs it through an ID Type Registry to mathematically validate its format.
    - It runs a read-only database query to verify that the ID actually exists and belongs to a customer the employee is authorized to view.
    - Once it finds exactly one match, it deterministically binds the ID to the payload, completely ignoring the AI's earlier guess.
4. **Layer 4: Policy and authorization**
    - The policy-as-code engine verifies that the authenticated employee has the correct permissions to perform a loan default inquiry.
    - It evaluates the risk tier, recognizing this as a read-only request rather than an irreversible write.
5. **Layer 5: Plan verifier and dry run**
    - The system checks state preconditions to ensure the account in question is actually an active loan account.
    - It uses Finacle's validation endpoints to simulate the inquiry safely before proceeding.
6. **Layer 6: Confirmation gate**
    - The employee is presented with a strictly factual summary pulled directly from the database (e.g., *"Confirm default check for Loan Account ending in 5544, registered to John Doe?"*).
    - Upon employee confirmation, the system generates a cryptographically signed token that is bound to the exact hash of the payload.
7. **Layer 7: Signed execution gateway**

    - This gateway acts as the sole component with credentials to the Finacle core.
    - It verifies the cryptographic signature from Layer 6; because the signature is mathematically valid, it allows the payload to pass through and retrieve the loan default data.
8. **Layer 8: Audit and telemetry**
    - The entire lifecycle of the query—including the masked ID, the database resolution, the policy approval, and the final execution—is recorded via OpenTelemetry traces.
    - This append-only log provides total explainability and ensures compliance with regional audit requirements.